

Functional Requirements:

Ticket Intake and Categorization:

- System should automatically collect tickets from multiple channels (email, chat, web forms, social media, etc.).
- Use Natural Language Processing (NLP) to classify tickets by issue type, urgency, and sentiment.
- Support custom category mapping defined by admins.

Assignment Engine:

- Assign tickets automatically based on:
 - Agent skill set.
 - Current workload (number and priority of open tickets).
 - Ticket priority and SLA.
 - Agent availability and working hours.
- Provide options for manual override by managers.

Prioritization:

- Prioritize tickets based on business rules (e.g., VIP customers, SLA deadlines).
- Update priority dynamically if issue escalates.

Notifications and Alerts:

- Notify agents when a new ticket is assigned.
- Alert managers if tickets remain unassigned for a threshold time.

Reporting and Analytics:

- Track metrics such as:
 - Average assignment time.
 - Agent utilization.
 - SLA compliance.
 - Ticket resolution trends.
- Generate dashboards for performance monitoring.

Integrations:

- Integrate with major ticketing systems (e.g., Zendesk, Jira, ServiceNow).

- Allow integration with internal HR/CRM systems for agent data.
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Non-Functional Requirements:

Category	Requirement
Performance	System should assign tickets within 5 seconds of creation.
Scalability	Must handle up to 10,000 tickets per hour without performance degradation.
Security	Ensure role-based access control and data encryption.
Reliability	System uptime of at least 99.5%.
Usability	Interface should be intuitive for agents and managers.
Maintainability	Support configuration of rules and workflows without coding.
Interoperability	API-based integration with third-party systems.

Constraints and Assumptions:

- Existing ticketing tools will provide accessible APIs.
 - All agents have defined skill sets and schedules in the system.
 - Organizational policies on data access and security are in place.
 - Internet connectivity is reliable for all users.
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Success Criteria:

- Reduction in average ticket assignment time by $\geq 50\%$.
 - Improvement in first response time and SLA adherence.
 - Increase in agent satisfaction (based on workload balance metrics).
 - Enhanced customer satisfaction scores (CSAT, NPS).
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Future Enhancements:

- Predictive ticket routing using machine learning.

- Sentiment-based escalation.
- Self-learning models for improving classification accuracy.
- Integration with workforce management for predictive staffing.

STREAMLINING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS



REQUIREMENT LIST

* Automated Routing:

- Keyword Recognition
- Rule-Based Conditions
- Load Balancing

* Agent Skill Matching:

- Expertise-Based Assignment
- Language Proficiency
- SLA Adherence Tier
- Urgency Level
- Availability Status

* Priorization Logic:

* Integration Capabilities:

- CRM Systems
- Chat Platforms
- Resolution Rates
- Agent Workload

* User Interface:

- Dashboard Time Metrics
- Resolution Rates
- Real-time Updates



NEW TICKET



AGENT SKILL MATCHING



ASSIGNED TICKET